Numerical Differentiation

MATH-151: Mathematical Algorithms in Matlab

October 2, 2023



A Refresher on the Derivative

- Since we learned about numerical integration last week, lets now look at the other primary focus of calculus, the derivative. Which at some point x is defined as

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  - The rate of change of the position of an object over time gives us its velocity. The rate of change of that is its acceleration
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- Derivatives are very informative, but very hard to do so we need to come up with approximations

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- These sure are straight-forward to do, but how good are they?

Finite Difference Error

- We can use the Taylor series approximations to analyze our error for the forward difference equation

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- The red part is our error, we describe it based on the power of h, so we say this is an $\mathcal{O}(h)$ method.
  - This means if we make h smaller, we should expect our error to also decrease proportional to h
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- Doing the same thing with the backward finite difference equation will give similar results.

Central Difference Method

- Since we know how to find error now, lets see if we can use $f(x)$, $f(x+h)$, and $f(x-h)$ to get a more accurate estimate that is $\mathcal{O}(h^2)$

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$\Rightarrow$ We Want $a+b+c=0$ *(This makes sure $f(x)$ terms cancel out)*

$$a - c = \frac{1}{h} \qquad \textit{(This keeps the } f'(x) \textit{ part)}$$

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- Which has error $\mathcal{O}(h^2)$ like we wanted. So now making h 10 times smaller will make your error 100 times smaller!

Second Order Derivatives

- Lets also see if we can use those same points to get an estimate of the second derivative

$$f''(x) \approx af(x+h) + bf(x) + cf(x-h)$$
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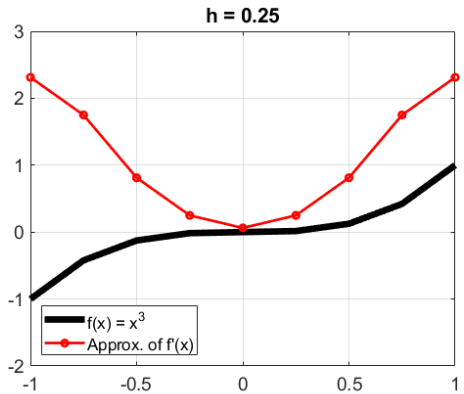
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- If we look further into the error, we see this is also a $\mathcal{O}(h^2)$ method

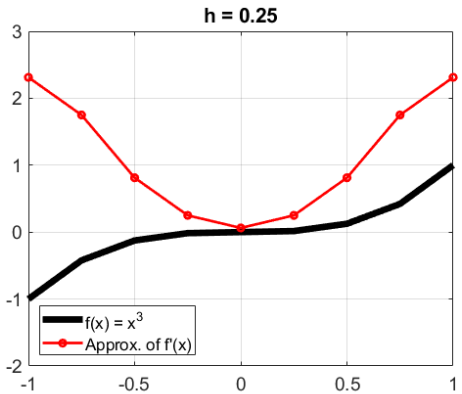
Derivative Functions

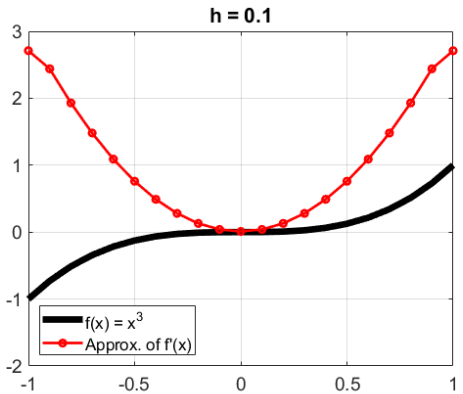
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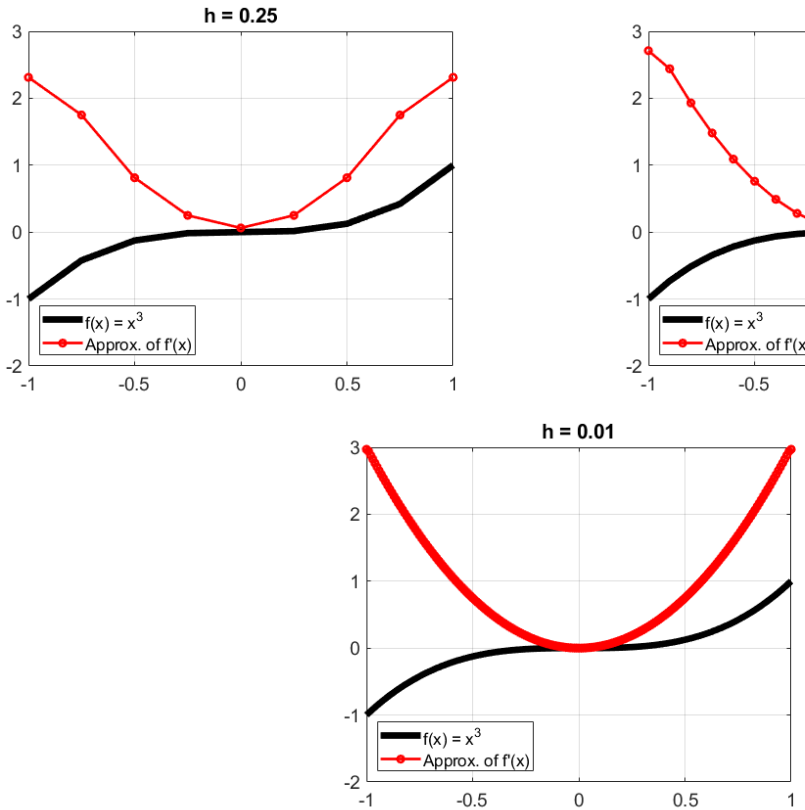
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- We can see clearly that more points gives us a more accurate estimate

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- Based on this information, Google thinks there is a 68.8% chance I was on a run, and a 17.9% chance of cycling. Heck there is a 0.01% chance I was flying!


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··"startLocation"·:·{LF
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····"longitudeE7"·:·-712869138,LF
····"sourceInfo"·:·{LF
····  "deviceTag"·:·1464504728LF
····}LF
··},LF
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- I was actually rollerblading, so good guess!


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